Comparing Temporal Logic Reward Shaping Approaches in Reinforcement Learning

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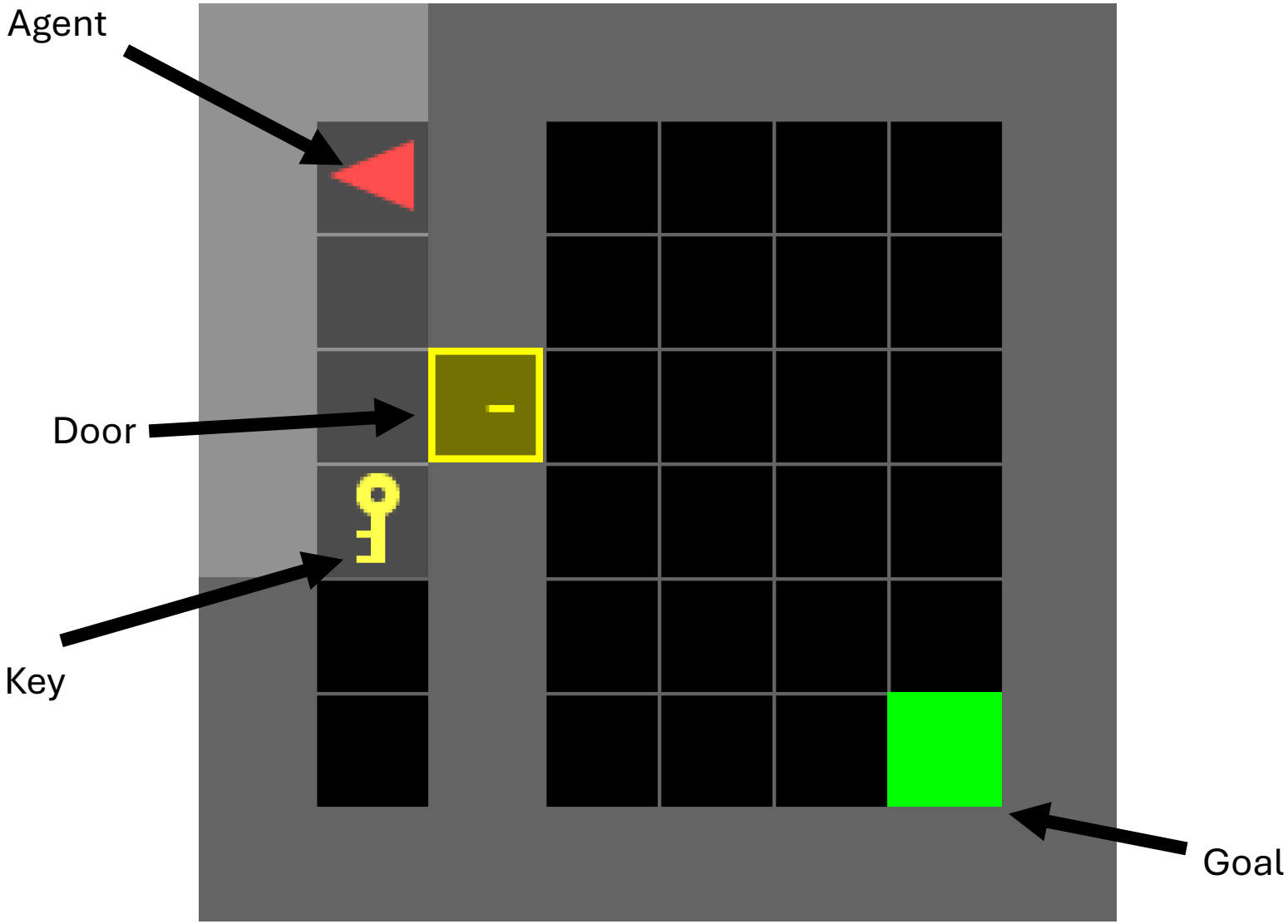
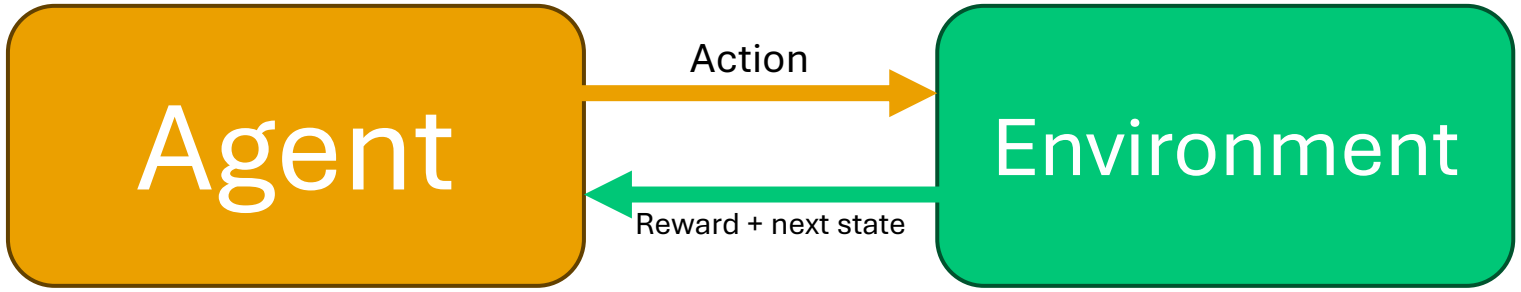
How do you reward behavior this complex?



The agent must get the key, unlock the door, then reach the goal, all in that order.

Reinforcement Learning



Agents learn entirely from **scalar reward signals**

- No explicit instructions… only feedback after acting

Sparse rewards fail on multi-step tasks

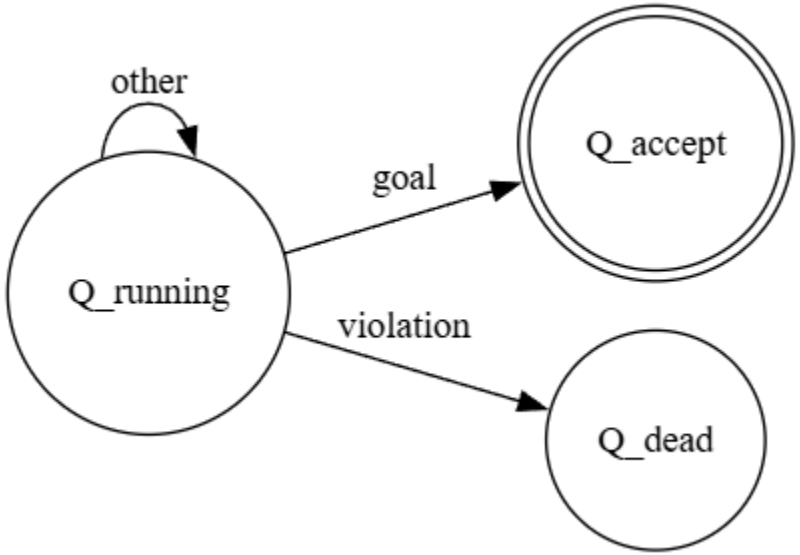
- Agent must accidentally complete the full sequence to get any signal

This is the core problem that **temporal logic** aims to solve

- Replace hand-crafted scalar rewards with formal behavioral specifications

Temporal Logic

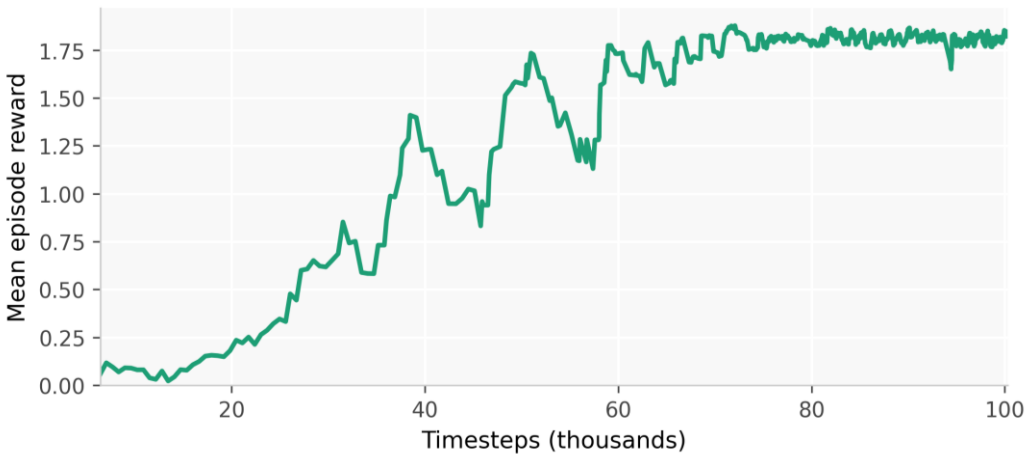
LTL – infinite horizon



Reward: +1 on accept, -1 on violation
Signal only at terminal states

Sparse Reward

TLTL – finite horizon



Reward: change in robustness each timestep

Dense Reward

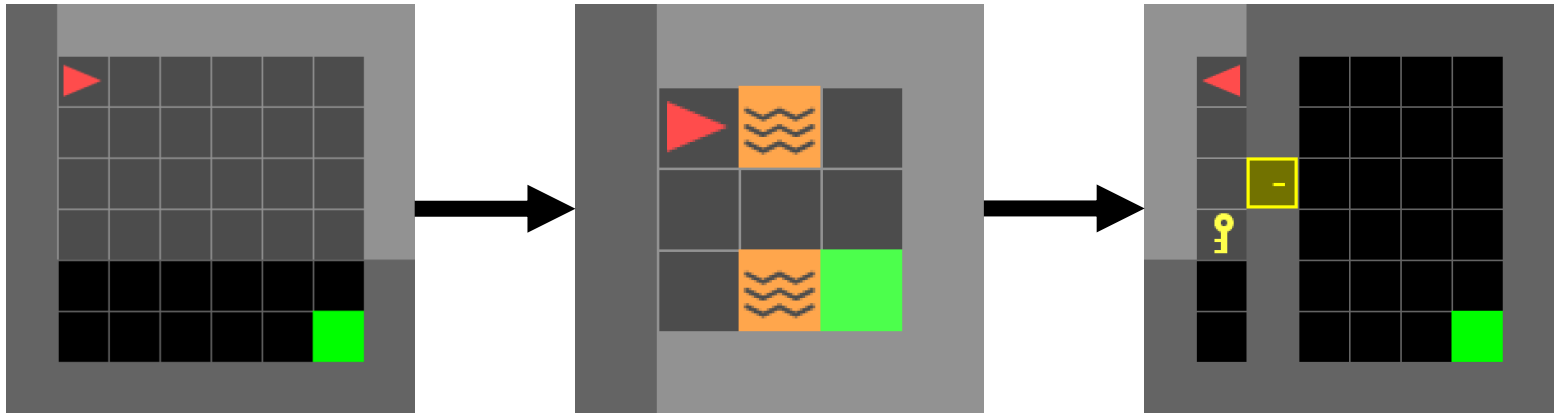
Specifications used in this study

F(goal)
Empty-8x8

G(¬lava) ∧ F(goal)
LavaGap-S5

F(key ∧ F(door ∧ F(goal)))
DoorKey-8x8

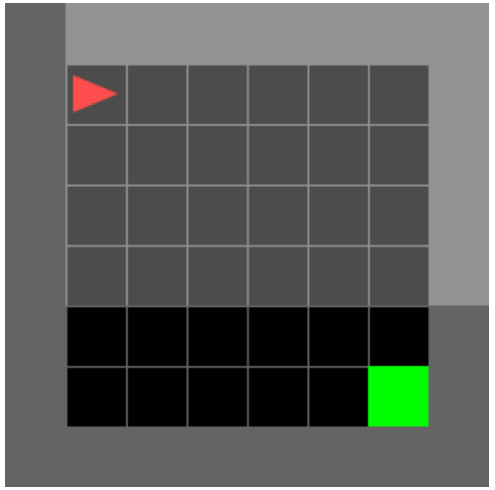
LTL and **TLTL** have each been studied independently, but which works better, and does **task complexity** change that answer?



Empty-8x8 LavaGap-S5 DoorKey-8x8

Increasing complexity

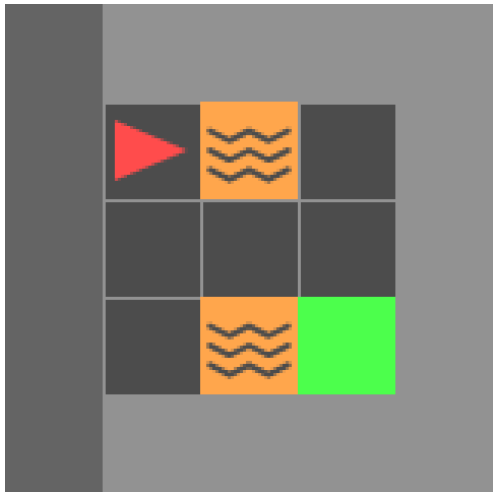
Empty-8x8



F(goal)

Basic traversal

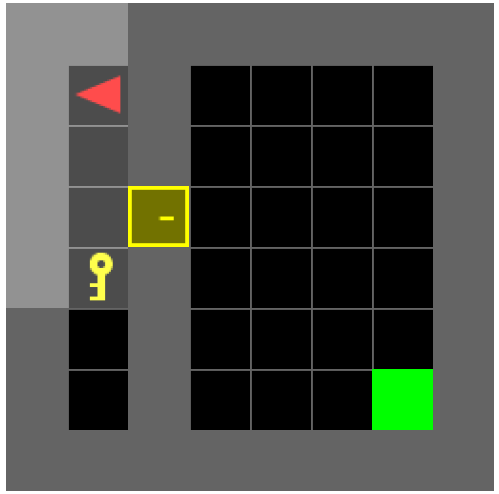
LavaGap-S5



G(¬lava) ∧ F(goal)

Hazard Avoidance

DoorKey-8x8



F(key ∧ F(door ∧ F(goal)))

Ordered subtasks

Three conditions tested on each environment

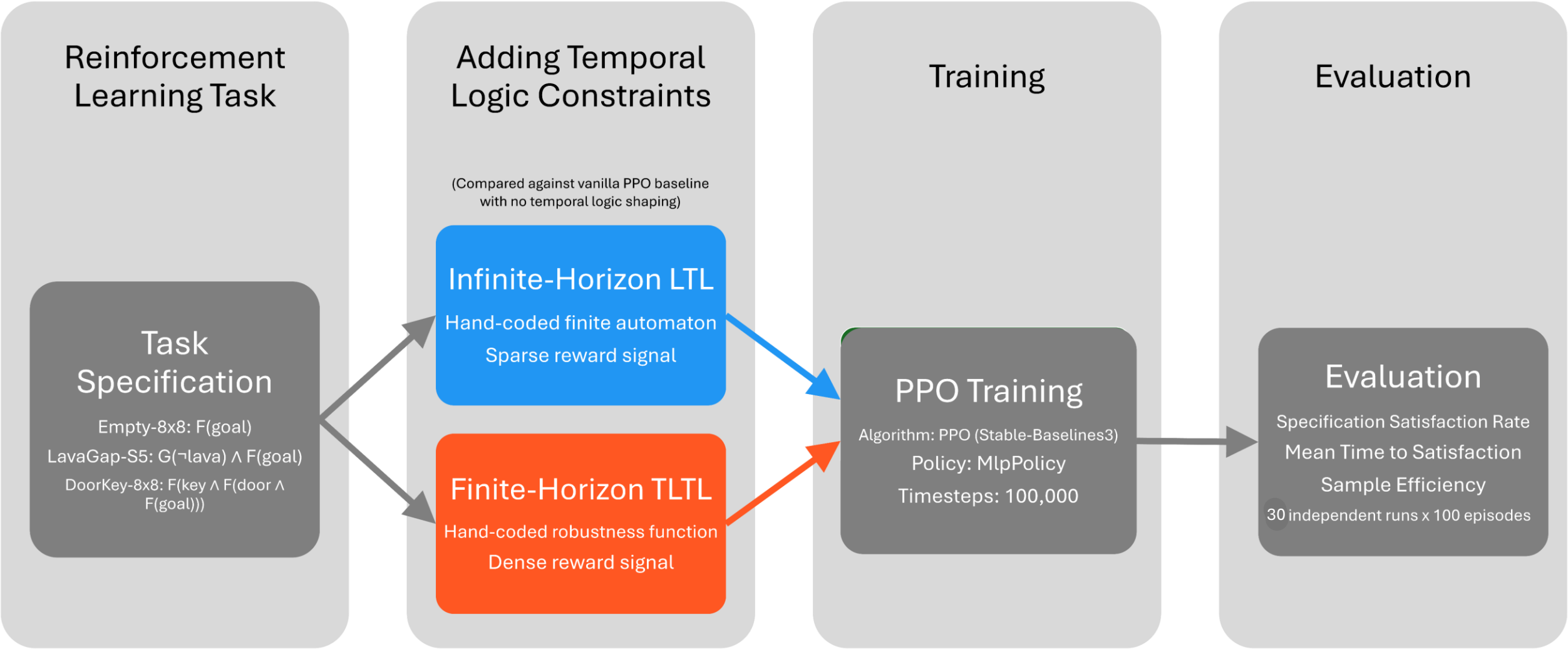
Baseline
Vanilla PPO, no shaping

LTL
Automaton wrapper

TLTL
Robustness function

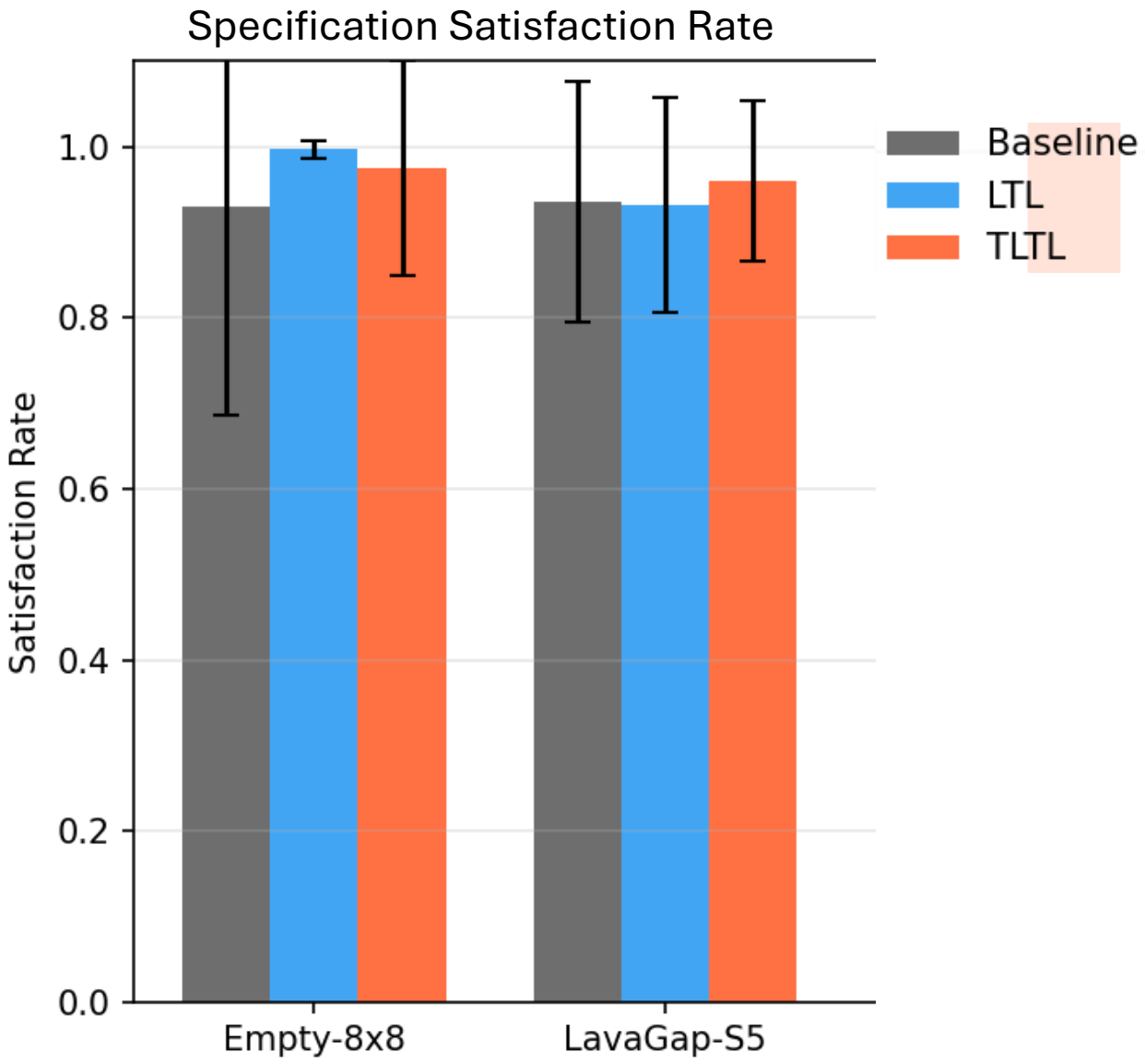
PPO, MlpPolicy, 100,000 timesteps, lr 3×10^{-4}, 30 independent runs, 100 eval episodes per run
identical hyperparameters across all conditions

Experiment Pipeline

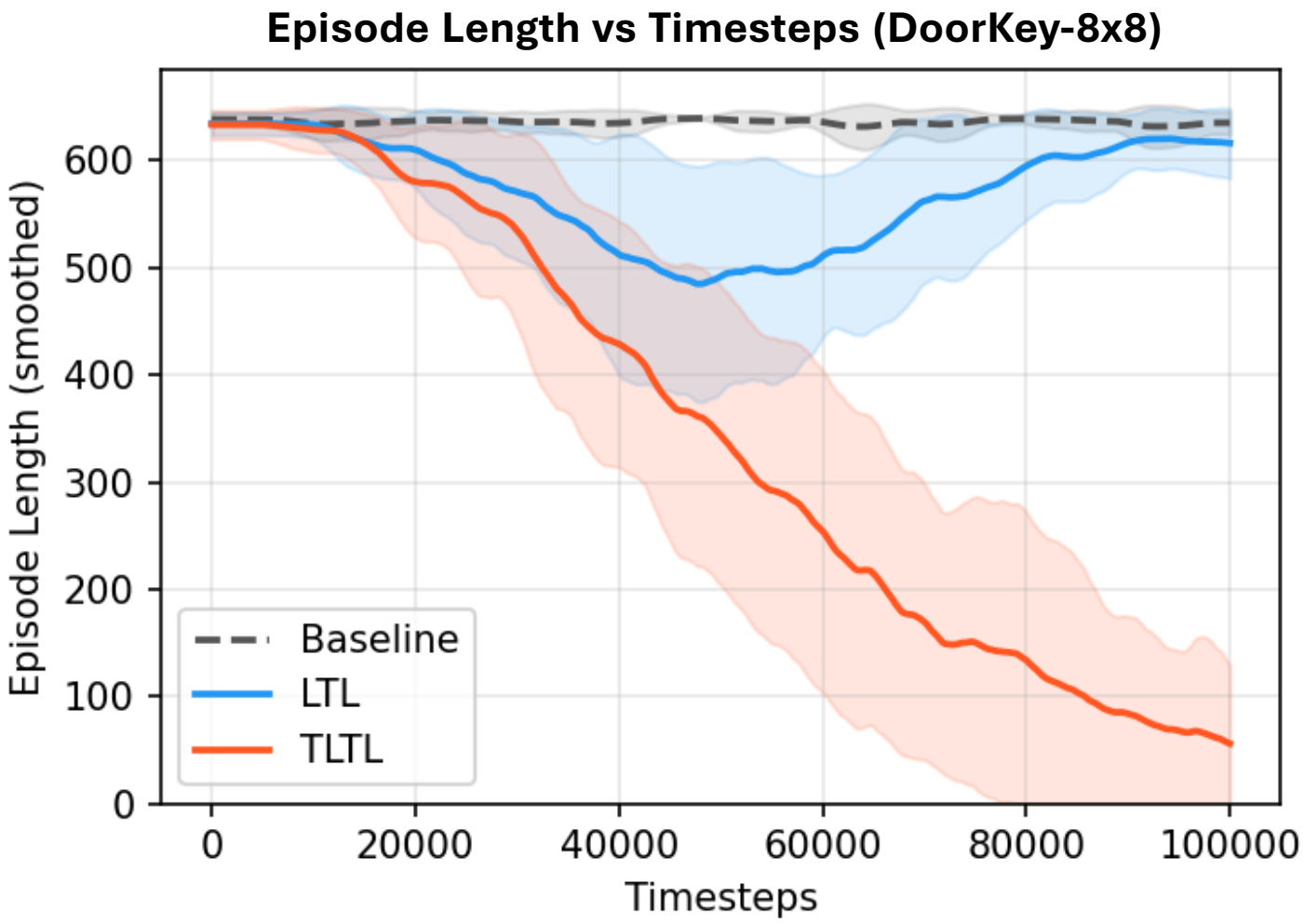


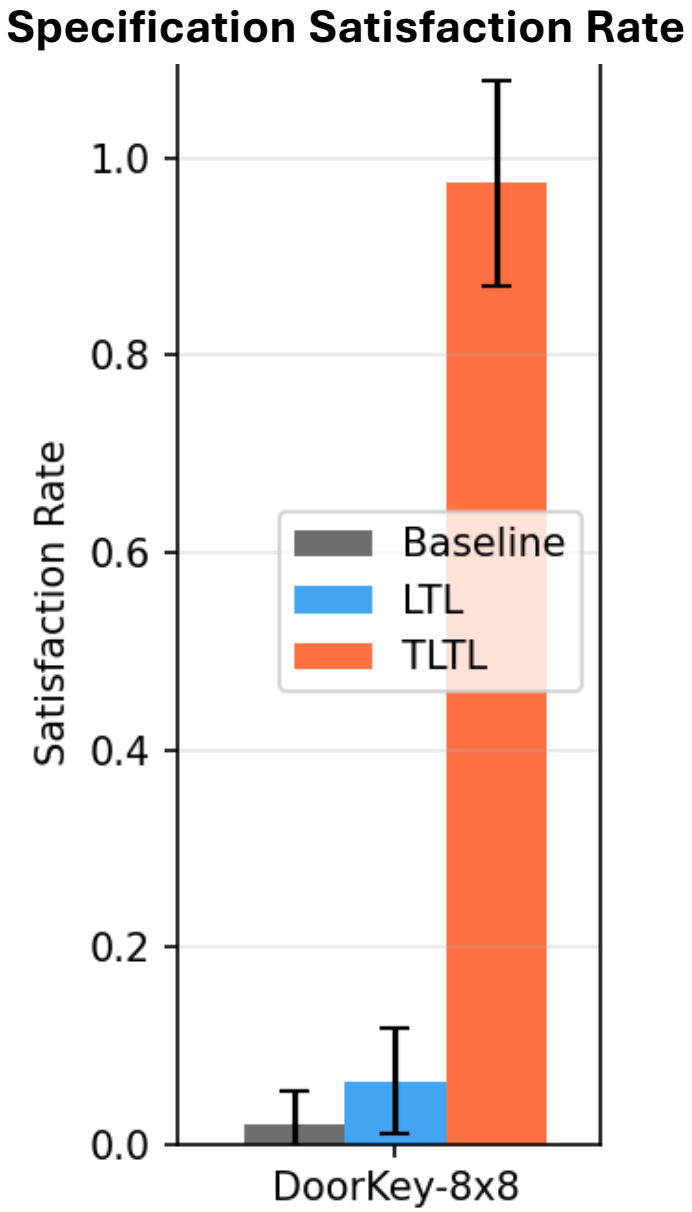
Results: Empty-8x8, LavaGap-S5

- On simpler tasks (Empty-8x8 and LavaGap-S5), differences are not statistically significant.



Results: DoorKey-8x8





Discussion

Why LTL struggled

- No signal until task completion
- Ordered subtasks rarely completed by chance
- Higher variance even on simpler tasks

Why TLTL succeeded

- Feedback at every timestep
- Partial credit for progress
- Dense signal guides ordered subtasks

Tradeoff

LTL - formal guarantees
Provable satisfaction properties, stronger theoretical foundation

TLTL - practical performance
Dense rewards, faster convergence, better on complex tasks

Neither is universally better, the right choice depends on your priority

Conclusions and Future Work

Conclusions

- TLTL significantly outperforms LTL and Baseline on DoorKey-8x8
- All methods comparable on simpler tasks
- Dense rewards are more practical for ordered subtask problems
- LTL showed higher variance even on simpler tasks

Future Work

- Evaluate across a wider range of environments
- Systematically vary hyperparameters to understand how they impact results